Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 2

Cloud Deployment and Service Models

IaaS, PaaS and SaaS, public, private, hybrid and multi-cloud, and an overview of AWS services.



Heng Rathpisey · Department GIC · Academic year 2025–2026

Recap

LAST SESSION

What cloud computing is

The on-demand delivery of computing services over the internet.

LAST SESSION

Five essential characteristics

On-demand self-service, broad network access, resource pooling, rapid elasticity and measured service.

LAST SESSION

Benefits and challenges

Cost, scalability and speed on one side. Cost control, security and vendor lock-in on the other.

Learning objectives

1. Describe the main cloud service models: IaaS, PaaS and SaaS.
2. Describe the main cloud deployment models: public, private, hybrid and multi-cloud.
3. Compare the models and map applications to them.
4. Name the main groups of AWS services.

THIS WEEK

Topic 2

Cloud deployment models and service models. Assessed by class participation and a quiz (CLO1).

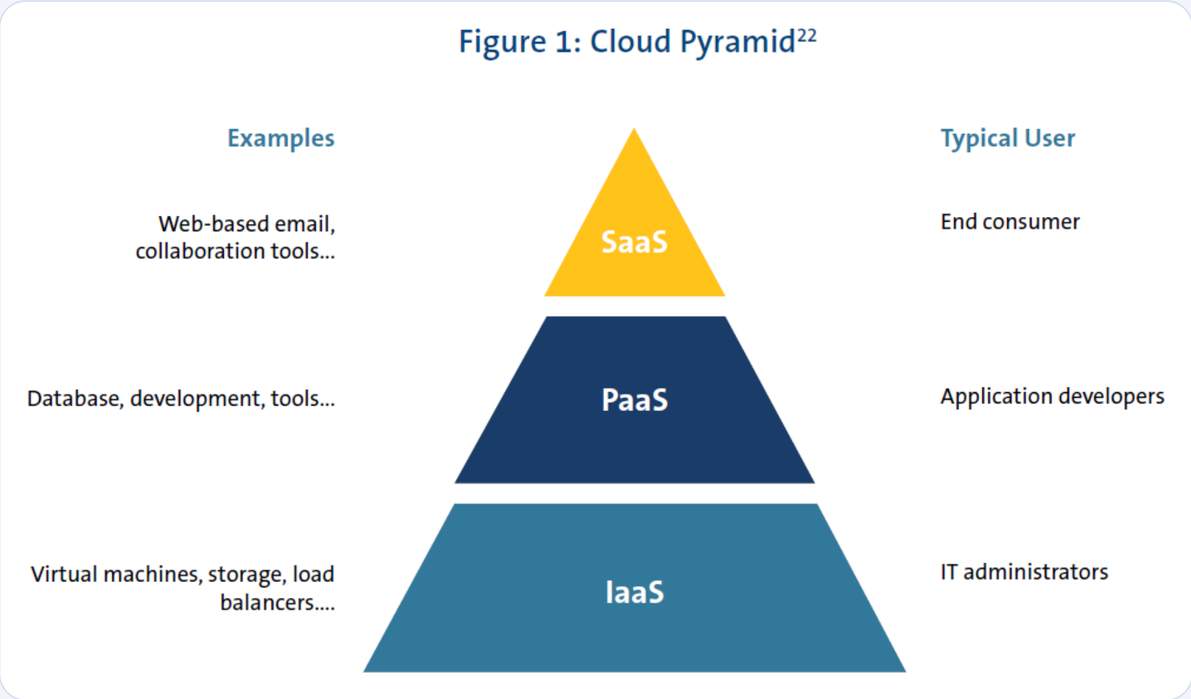
PART 1

Cloud service models

How much of the stack the provider runs for you, and how much stays with you.

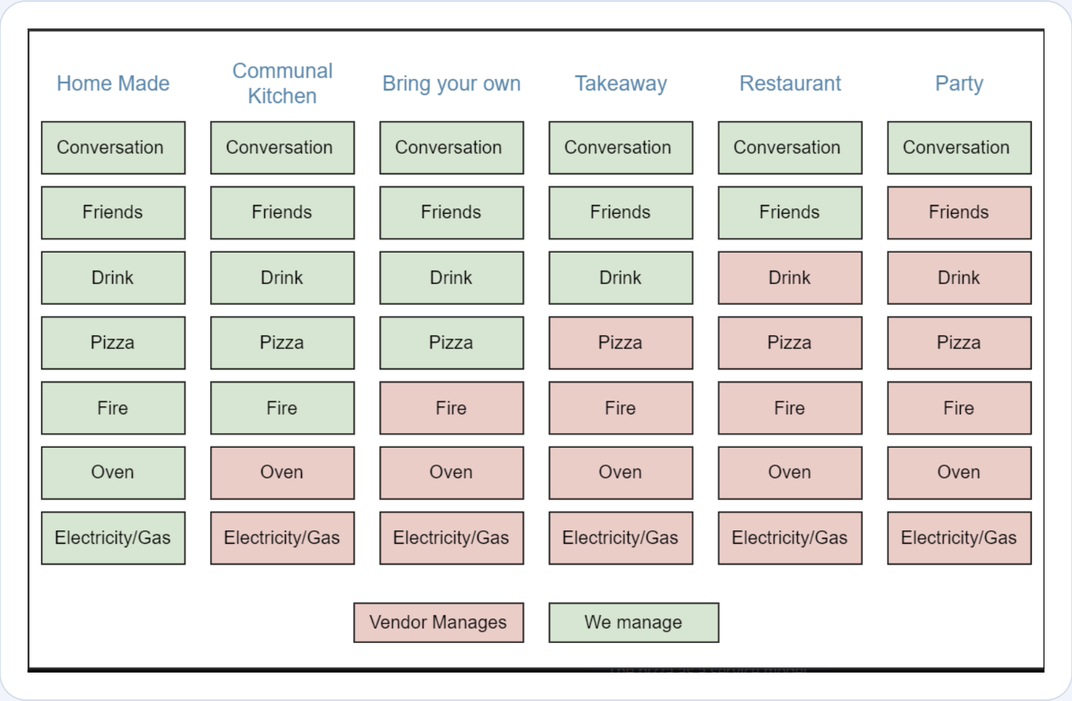
Week 2

Cloud service models



Three layers, three audiences: IaaS for IT administrators, PaaS for application developers, SaaS for end consumers.

Analogy: pizza as a service



From home made to a party, we manage less and the vendor manages more. Cloud service models work the same way.

Who manages what?

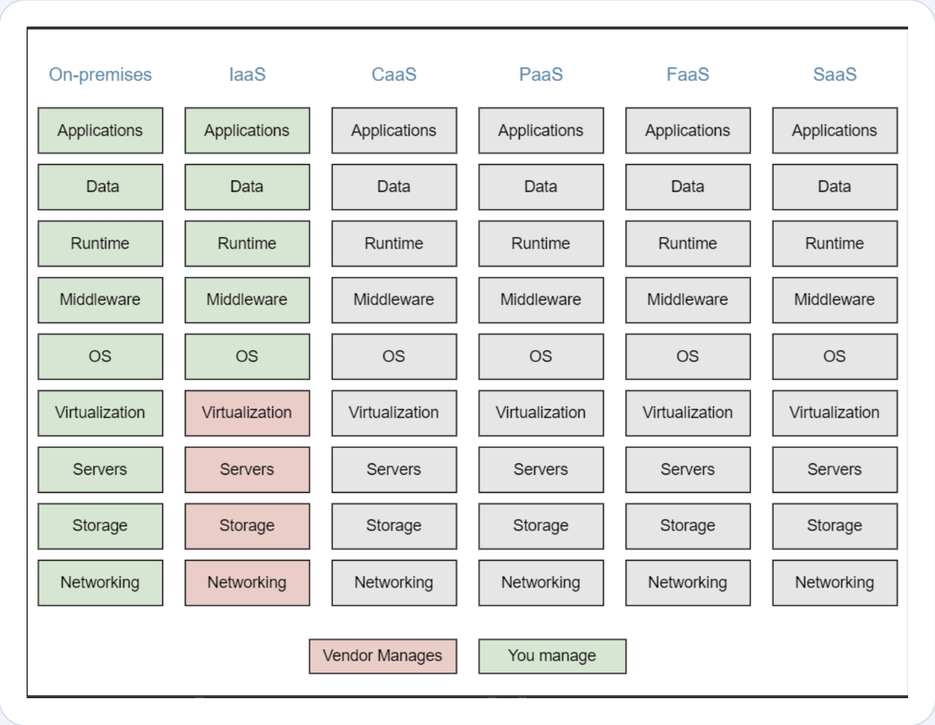
Layer	On-premises	IaaS	PaaS	SaaS
Applications and data	You	You	You	Vendor
Runtime and middleware	You	You	Vendor	Vendor
Operating system	You	You	Vendor	Vendor
Virtualization, servers, storage, networking	You	Vendor	Vendor	Vendor

The further right, the less you manage and the less you control. In SaaS, you still own the data you put in.

Infrastructure as a Service (IaaS)

The vendor provides the *infrastructure*. You run everything on top of it.

The vendor manages virtualization, servers, storage and networking. You manage the operating system, middleware, runtime, data and applications.



When to use IaaS?

Lift and shift

We move our workload from physical servers to virtual machines set up in the cloud.

Cloud advantages

We take advantage of the platform's scalability, high availability, and single control and management plane.

Who it targets

IT professionals or admins who are primarily responsible for setting up the infrastructure and managing it.

IaaS services by cloud platform

Cloud platform	Services
AWS	Amazon Elastic Compute Cloud (EC2) and Amazon Simple Storage Service (S3)
Microsoft Azure	Azure Virtual Machines, Virtual Machine Scale Sets and Azure Virtual Networks
Google Cloud Platform	Google Compute Engine and Google Cloud Storage

IaaS: advantages and drawbacks

Advantages

- Virtual machines replace physical hardware
- The ability to quickly scale up or down
- Pay only for what we use
- Suitable for high-performance computing workloads

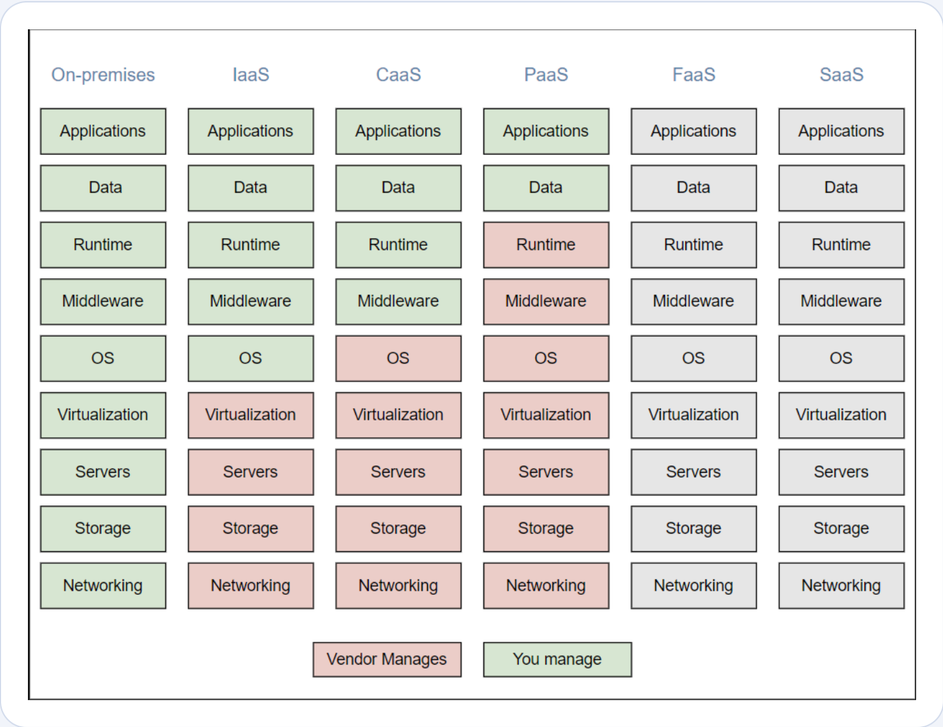
Drawbacks

- Operations management: we still patch, secure and run the servers
- Expensive, if machines are left running or oversized

Platform as a Service (PaaS)

PaaS provides the *development tools and frameworks* to build cloud applications rapidly.

We deploy our code and data. The vendor also manages the operating system, middleware and runtime.



PaaS services by cloud platform

Cloud platform	Services
AWS	AWS Elastic Beanstalk and AWS Lambda
Microsoft Azure	Azure Logic Apps, Azure App Service, Azure Functions and Azure Cosmos DB
Google Cloud Platform	Google App Engine and Google Cloud Run

AWS Lambda and Azure Functions run single functions on demand. They are often given their own name: Function as a Service (FaaS).

PaaS: advantages and drawbacks

Advantages

- No servers or operating systems to patch
- Faster from code to a running application
- Scaling is built into the platform

Drawbacks

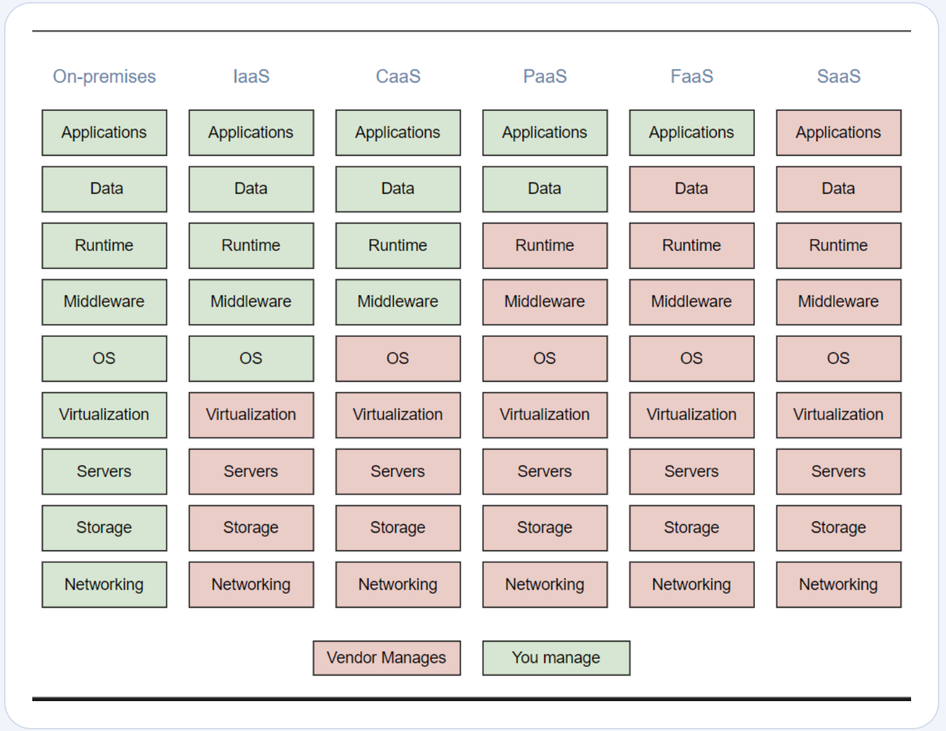
- Less control over the environment
- Limited to the languages and versions the platform supports
- Harder to move to another vendor

Software as a Service (SaaS)

SaaS is the delivery of *software over the internet*.

We are simply a consumer of the application, without concern about how it is developed, managed and operated.

A common way of delivering a SaaS product or service is via subscriptions.



SaaS examples

Company	SaaS products
Dropbox and Salesforce	Dropbox and Salesforce
Microsoft	Office 365
Google	Google Workspace and Gmail
Twitter	Twitter
Meta	Facebook, WhatsApp and Instagram

SaaS: advantages and drawbacks

Advantages

- Nothing to install, run or update
- Available from any device with a browser
- Predictable subscription cost

Drawbacks

- Least control and customization
- Our data lives with the vendor
- We depend on the vendor’s availability and roadmap

Two more columns: CaaS and FaaS

BETWEEN IAAS AND PAAS

Container as a Service (CaaS)

Sits between IaaS and PaaS. We package the application as containers, and the vendor runs and orchestrates them. On AWS: Amazon ECS and Amazon EKS.

BETWEEN PAAS AND SAAS

Function as a Service (FaaS)

Sits between PaaS and SaaS. We write single functions, and the vendor runs them only when an event calls them. On AWS: AWS Lambda.

Both appear in the responsibility pictures. We return to containers and serverless later in the course.

Service model comparison

	IaaS	PaaS	SaaS
Typical user	IT administrators	Application developers	End consumers
You get	Virtual machines, storage, load balancers	Databases, development tools, a runtime	A finished application
You manage	OS, runtime, data, applications	Data and applications	Only your data and settings
Control	Most	Medium	Least
Example	Amazon EC2	AWS Elastic Beanstalk	Gmail, Office 365

Which service model fits?

Moving 20 existing servers to the cloud

IaaS. Lift and shift onto virtual machines, keeping the same operating systems and software.

A student team building a web app

PaaS. Push the code and let the platform run, patch and scale it.

Email and shared documents for a school

SaaS. Subscribe to Google Workspace or Office 365, with nothing to operate.

Resizing a photo each time one is uploaded

FaaS. A function that runs only when the upload event arrives.

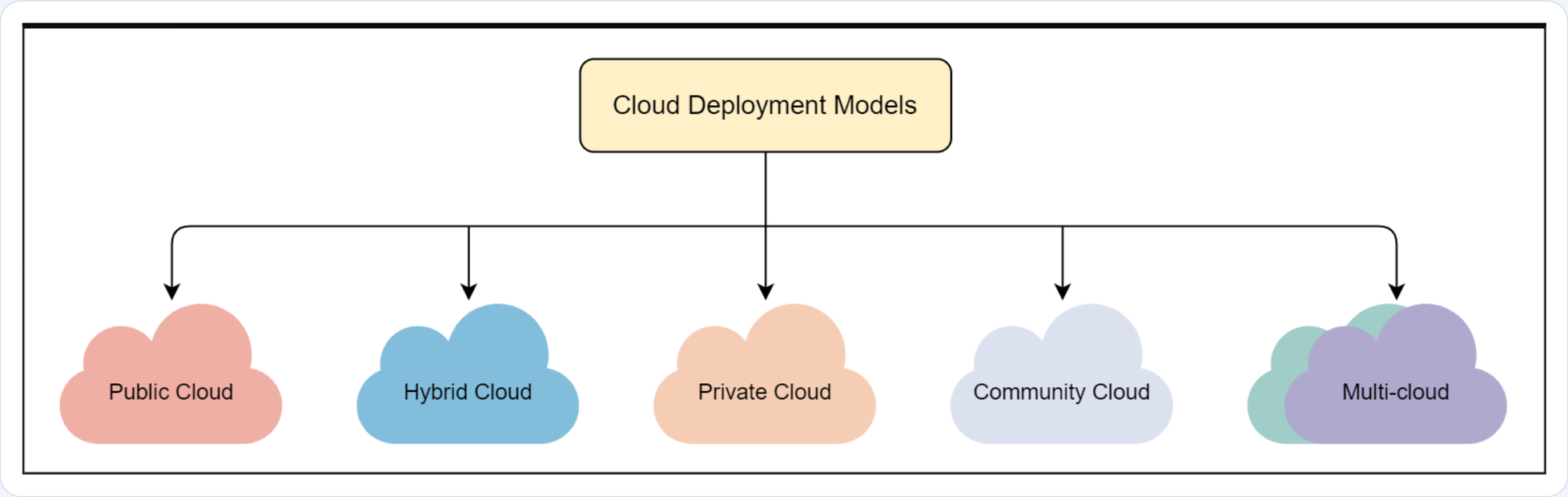
Ask first: how much do we want to manage ourselves?

PART 2

Cloud deployment models

Who owns the cloud, who shares it, and where it runs.

Cloud deployment models



NIST defines four: public, private, hybrid and community. Multi-cloud is the fifth pattern we meet in practice.

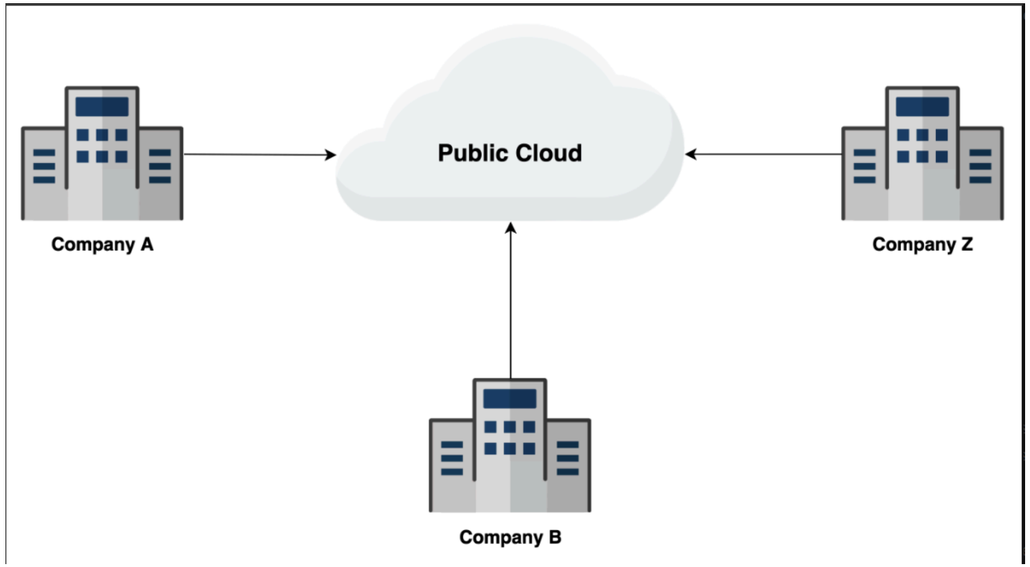
Source: NIST

Public cloud

Managed by *third-party providers*, and shared among many customers.

Computing resources such as storage and computing power are offered over the internet, like a utility service such as electricity.

Examples: Amazon Web Services, Microsoft Azure and Google Cloud Platform.



Public cloud: why it is widely used

Easy

Public clouds are widely used due to their ease of use and onboarding.

Cost-effective

The shared model serves a large customer base efficiently, and the savings are passed on to consumers.

Still secure

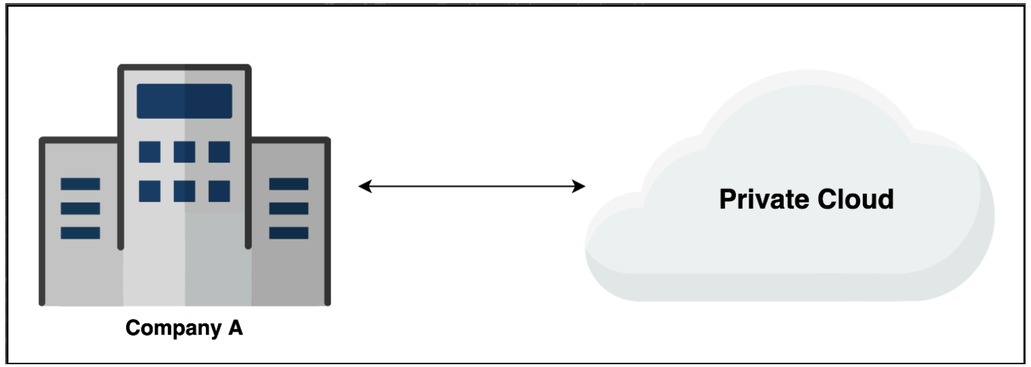
Access to data and resources is restricted to authorized users, even though the hardware is shared.

Private cloud

Exclusive to *one organization*, with dedicated resources.

Similar to traditional infrastructure, but with modern technologies like virtualization and automation for easier management.

Two types: hosted private clouds, provided by third-party providers, and internal private clouds, managed by the organization itself.



Private cloud: control at a cost

Control

Enhanced control over security, privacy and compliance.

Cloud capabilities

Still provides self-service, scalability and elasticity.

Cost

The dedicated environment costs more than public clouds. It is the most expensive model to operate.

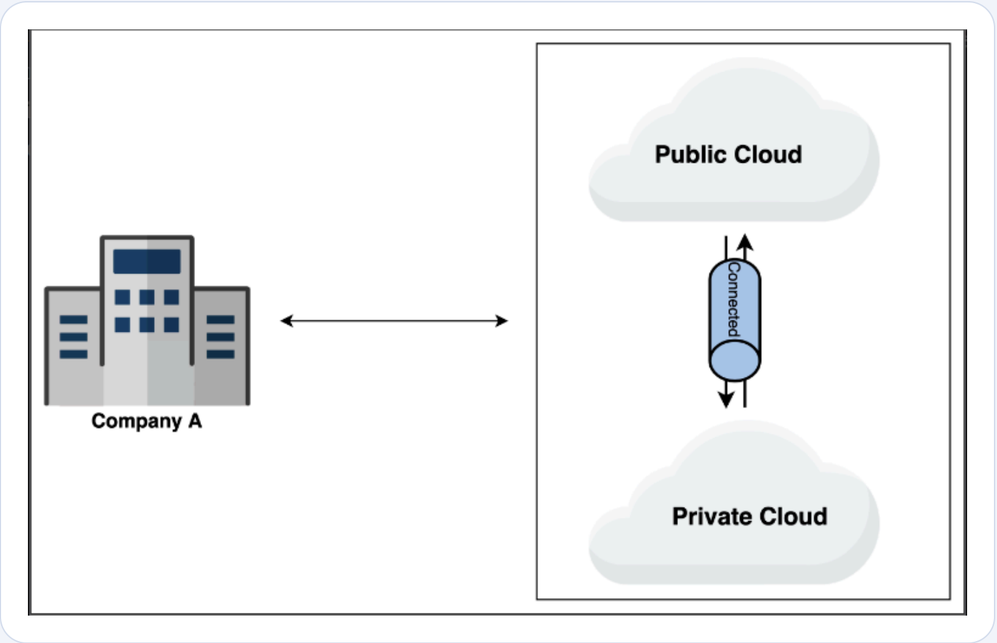
Examples: Microsoft Azure Stack and Red Hat OpenShift.

Hybrid cloud

A *combination* of public and private cloud.

Some resources are hosted publicly, while others remain on-premises. A common use is extending on-premises infrastructure to the cloud.

Data stays within private, internal networks, and services running in the public cloud get secure, controlled access to it.



Why hybrid is becoming more common

Existing investment

Many enterprises have invested heavily in on-premises applications that are critical and sensitive. A complete shift is impractical.

Mission-critical applications

On-premises systems often host applications that cannot be easily or securely moved to the cloud.

Own pace

Enterprises adopt cloud technologies gradually, at their own pace, while maintaining agility.

The price: complexity

Connecting on-premises environments to cloud platforms securely needs complex infrastructure.

Hybrid cloud solutions

Azure Stack and Azure Arc

Microsoft's solutions for extending Azure services and capabilities to on-premises environments.

AWS Outposts

A fully managed service that extends AWS infrastructure, services, APIs and tools to a datacenter or on-premises facility.

Google Anthos

An open platform to run an application anywhere, on existing on-premises hardware or in the public cloud.

For cloud vendors, hybrid is a strategic way to attract customers who are not ready for a full cloud transition.

Community cloud

Shared by *several organizations* with the same concerns.

Who shares it

A specific community with shared concerns, such as a mission, security requirements, policy or compliance.

Who runs it

One or more of the organizations in the community, a third party, or a combination.

Where it fits

For example, universities sharing a research cloud, or government agencies sharing one that meets their regulations.

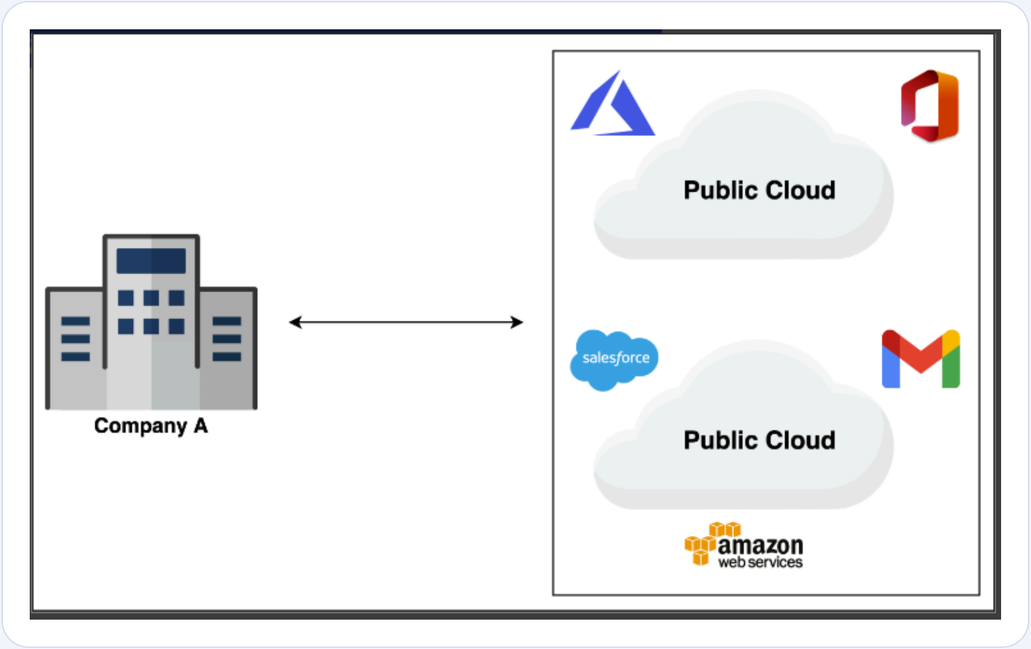
Source: NIST

Multi-cloud

Using cloud services from *more than one provider*.

Common usage: several SaaS applications at once, like Microsoft Office 365, Zoom or Salesforce.

In larger businesses, it usually means using PaaS or IaaS offerings from different providers such as AWS, GCP or Azure.



Multi-cloud: pros and cons

Pros

- It is reliable.
- It has low vendor lock-in.
- It is potentially cost-saving.

Cons

- It has high operational complexity.
- It has high latency, when traffic crosses between clouds.

Deployment models: pros and cons

	Public	Private	Hybrid	Multi-cloud
Cost	Low	Very high	Moderate	High
Security	Good	Very good	Moderate	Moderate
Complexity	Low	High	Moderate	Very high
Control	Low	High	Moderate	Moderate
Scalability	High	Moderate	High	High
Flexibility	Low	High	Moderate	Moderate

Which deployment model fits?

A startup launching a mobile app

Public cloud. No hardware to buy, low cost to start, and room to grow.

A bank with strict rules on customer data

Private cloud, or hybrid with the sensitive data kept in the private part.

A factory with on-site systems, adding analytics

Hybrid cloud. The line-of-business systems stay on-premises, and the cloud adds new capabilities.

A company using Office 365 and AWS

Multi-cloud. Different providers for different jobs.

Ask first: how sensitive is the data, and what do we already own?

PART 3

AWS services overview

The public cloud we use in this course, and the main groups of services it offers.

AWS services by category

Category	Services	What they are for
Compute	Amazon EC2, AWS Lambda, Amazon ECS and EKS	Running code and applications
Storage	Amazon S3, Amazon EBS, Amazon EFS	Keeping objects, disks and shared files
Networking	Amazon VPC, Amazon CloudFront	Private networks and content delivery
Database	Amazon RDS, Amazon DynamoDB	Relational and nonrelational data
Security	AWS IAM	Who can do what
Cost management	AWS Cost Explorer, AWS Budgets	Seeing and controlling spending

AWS services and the service models

01

IaaS

Amazon EC2, Amazon EBS and Amazon VPC. We get servers, disks and networks, and run the rest.

02

PaaS

AWS Elastic Beanstalk and Amazon RDS. We bring code or data, and AWS runs the platform.

03

FaaS

AWS Lambda. We bring one function, and AWS runs it when an event arrives.

AWS is a public cloud. With AWS Outposts it also reaches into hybrid setups.

How we use AWS in this course

AWS Academy Learner Lab

Each student creates and maintains an account, with a quota of 50 USD per account.

Stay within the quota

Monitor your AWS usage, and clean up the resources you create after each lab.

Lab reports

Complete the lab assignments and submit the lab reports on time.

Source: course specification

QUIZ

Post-knowledge quiz

Cloud service models and cloud deployment models.

Next session



01

Cloud infrastructure



02

Virtualization and compute



03

Amazon EC2 and Lab 1